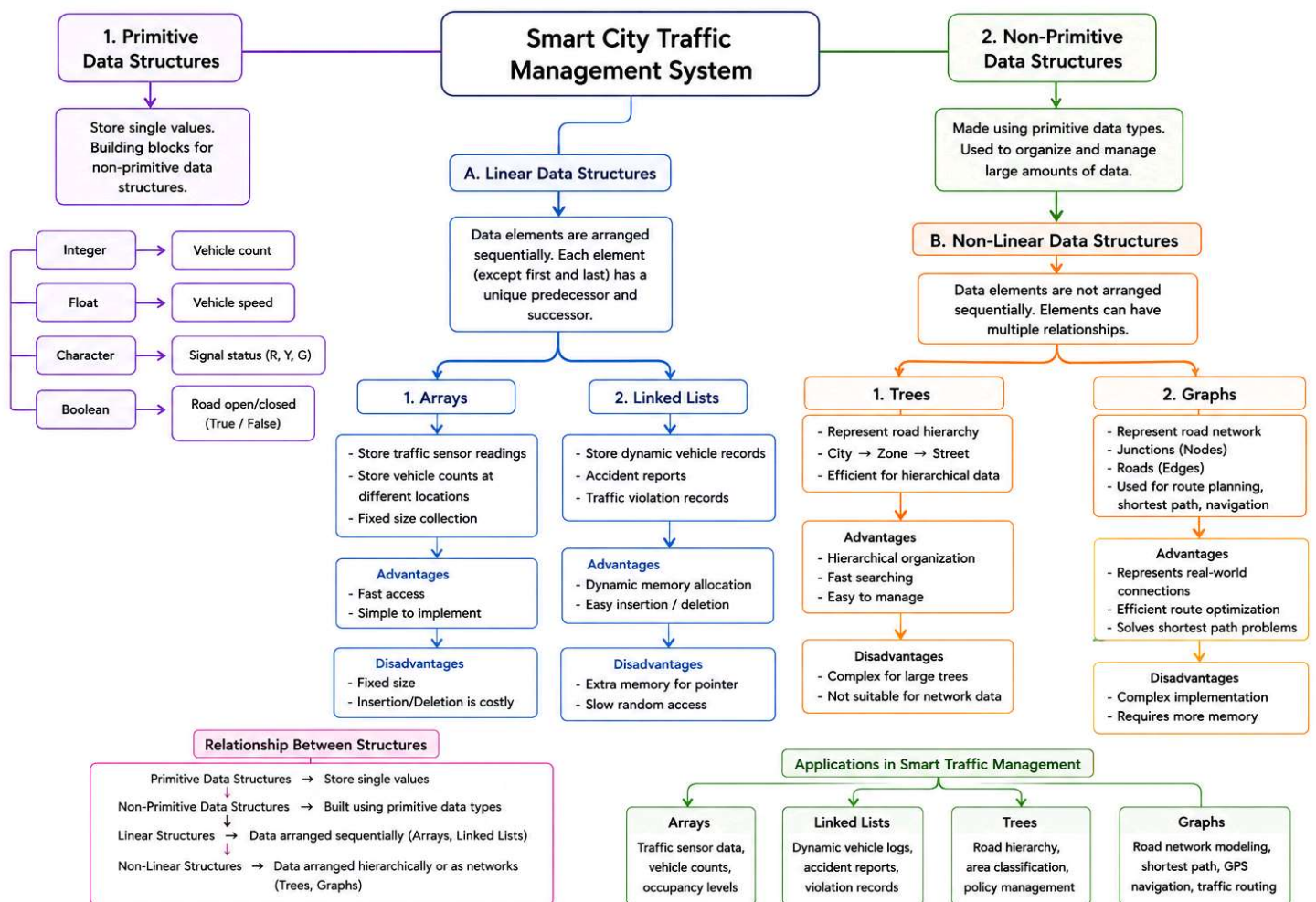




Ride-Sharing Application (Real-Time Driver Matching)

1. Algorithm

Linear
Search

Hash
Table

KD-Tree
(Best)

Increasing Input Size

- More Users
- More Drivers
- More Processing
- Slower if algorithm is poor

2. Input Size (n)

Users

Drivers

Locations

Best Algorithm

- KD-Tree
- Fast Search
- Efficient
- Real-time Matching

3. Time Complexity

$O(n)$

$O(\log n)$

Faster
Search

4. Space Complexity

Memory
Usage

Storage
Required

5. Efficiency

Fast
Response

Low
Memory

Scalable

6. Performance Measurement

Response
Time

Throughput
(matches/sec)

CPU &
Memory